NST21022 - Practical for Network Switching and Routing

Department of Information and Communication Technology

Faculty of Technology

Lab sheet :03
Reg. Number: SEU/IS/20/ICT/084
Academic Year :2020/2021
Practical No :03

Title:

- **Mac addresses and Routers**

Aim:

- **Get to know about MAC addresses and its important**
- **Get to know about Router and its important**

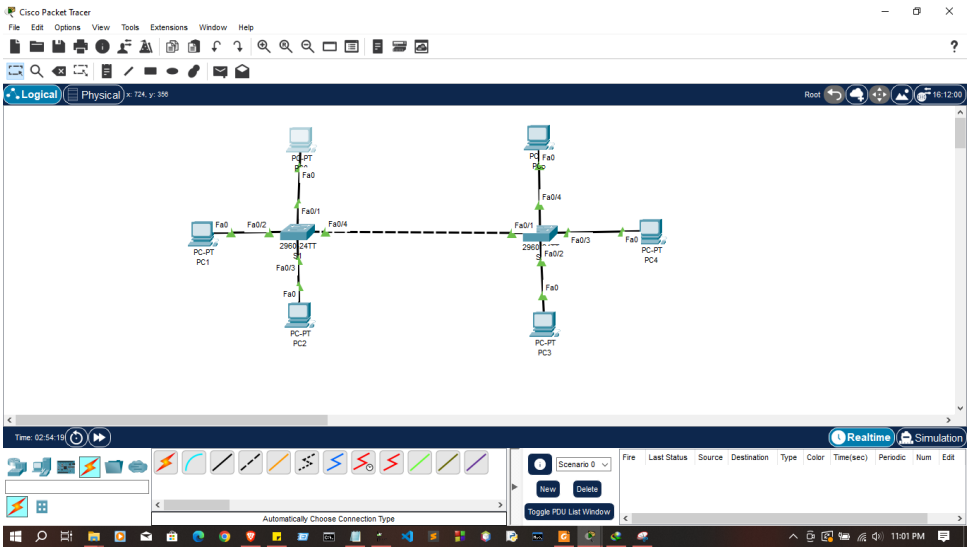
Task:

- **Create a simple network and check MAC addresses**
- **Attach a router to the network and configure and check**

<u>Activities</u>

Exercise 01:

1. Create a simple network with following devices and check MAC address and configure IP addresses.
    a. 2 Switches
    b. 6 PCs
    c. Use suitable connection



Switches 01

```
enable
configure terminal
hostname RegNo-S1
no ip domain-lookup
line console 0
password 1234
login
exit
enable secret class
service password-encryption
interface vlan 1
ip address 192.168.1.1 255.255.255.0
no shutdown
banner motd $

##################################################################
##################################################################

            Autherized user Only
                        Switche1

##################################################################
##################################################################
$
```

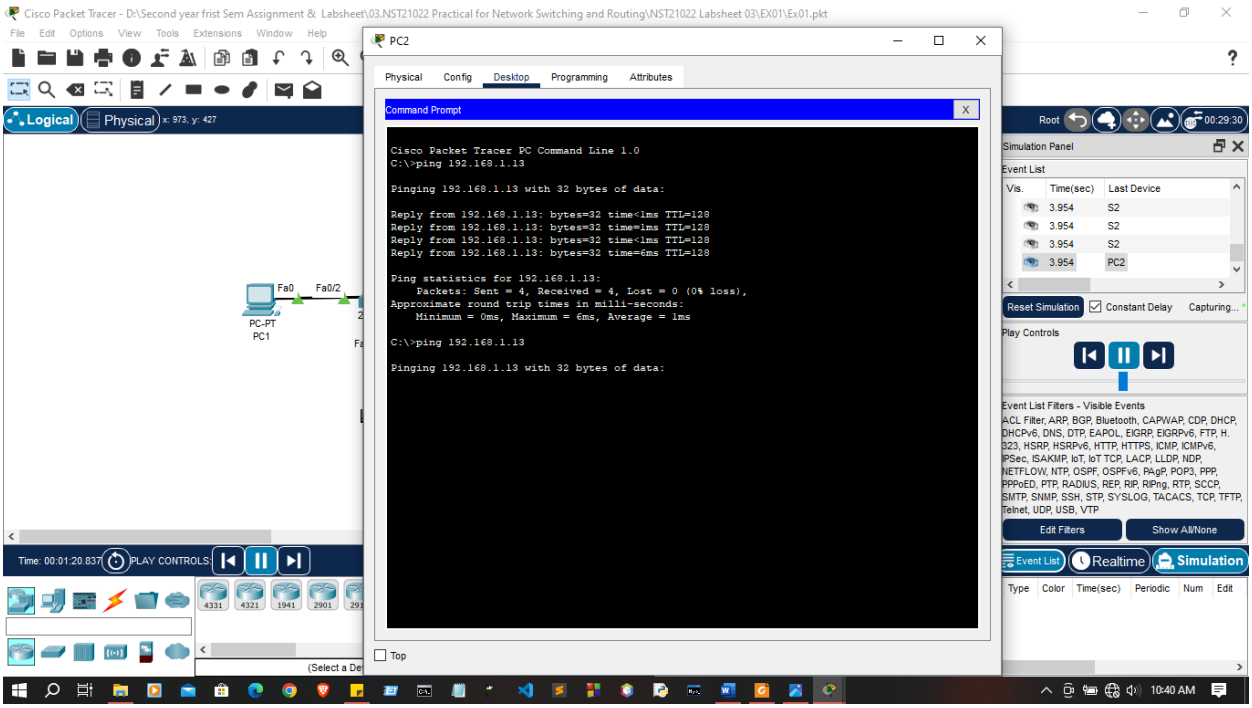
```
exit
exit
show running-config
copy running-config startup-config
enable
```

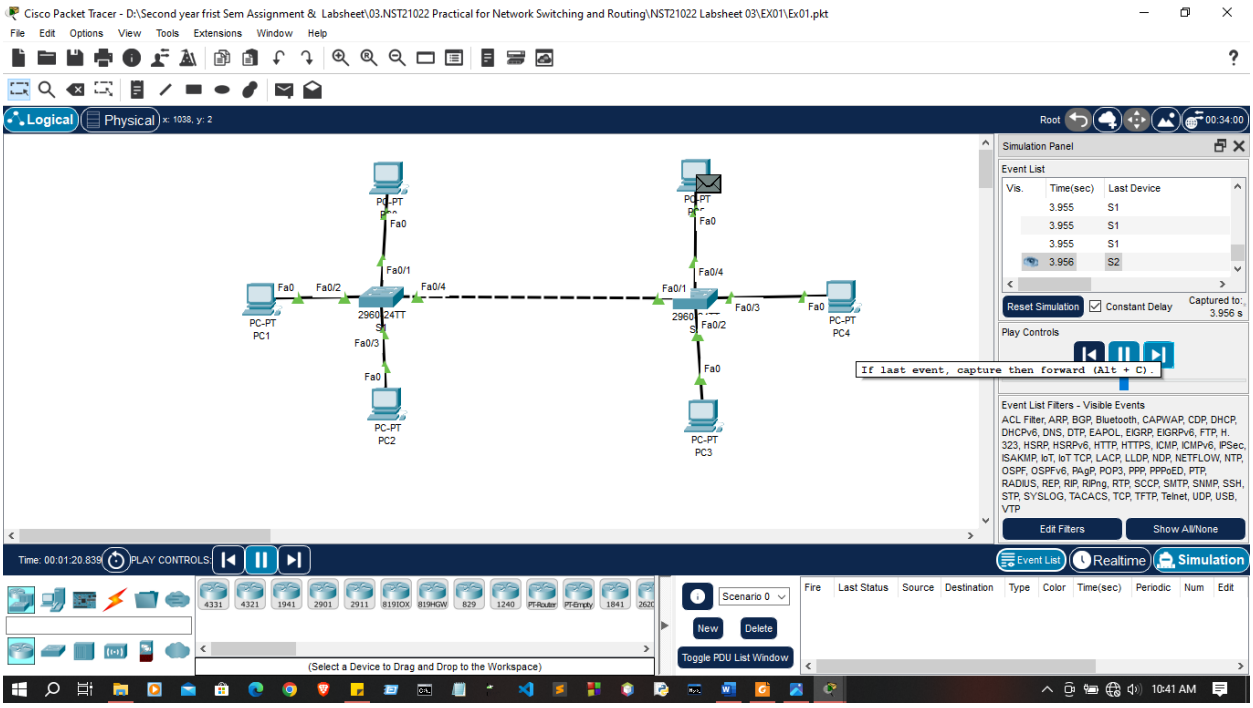
Switches 02

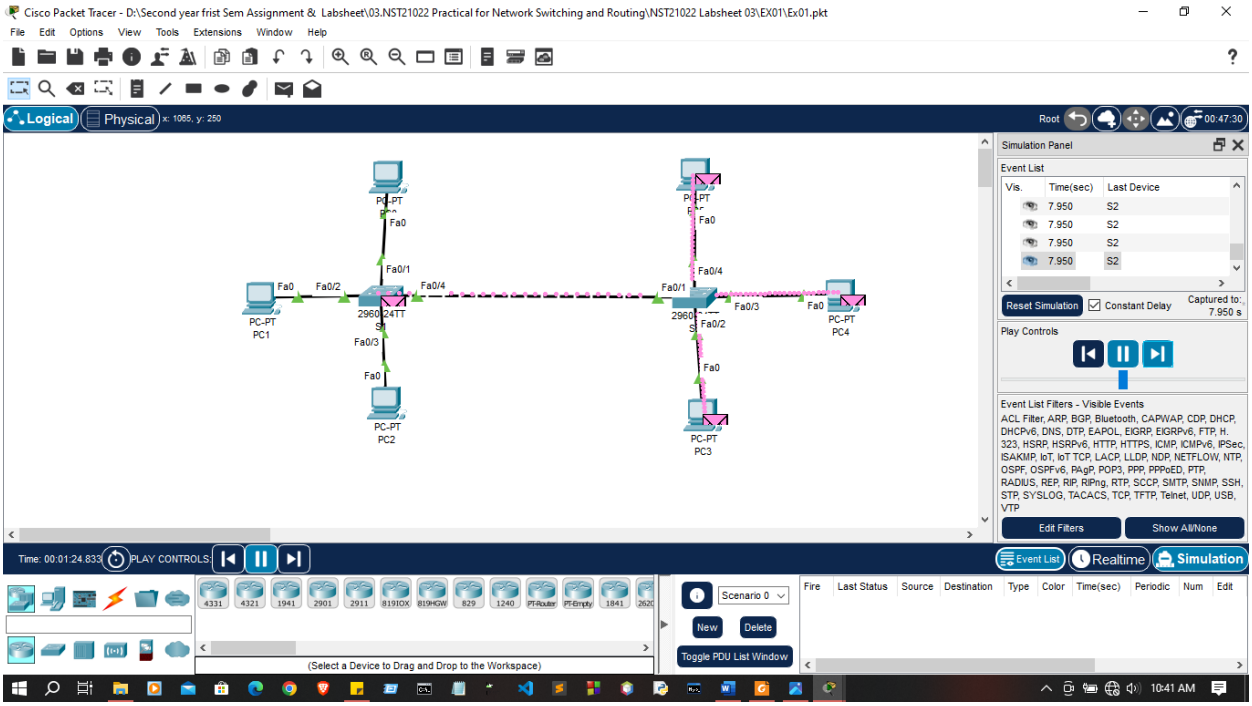
```
enable
configure terminal
hostname RegNo-S1
no ip domain-lookup
line console 0
password 1234
login
exit
enable secret class
service password-encryption
interface vlan 1
ip address 192.168.1.2 255.255.255.0
no shutdown
banner motd $

#################################################################
###############
#################################################################
###############

            Autherized user Only
                        Switche2
```

```
##################################################################
$

exit
exit
show running-config
copy running-config startup-config
enable
```

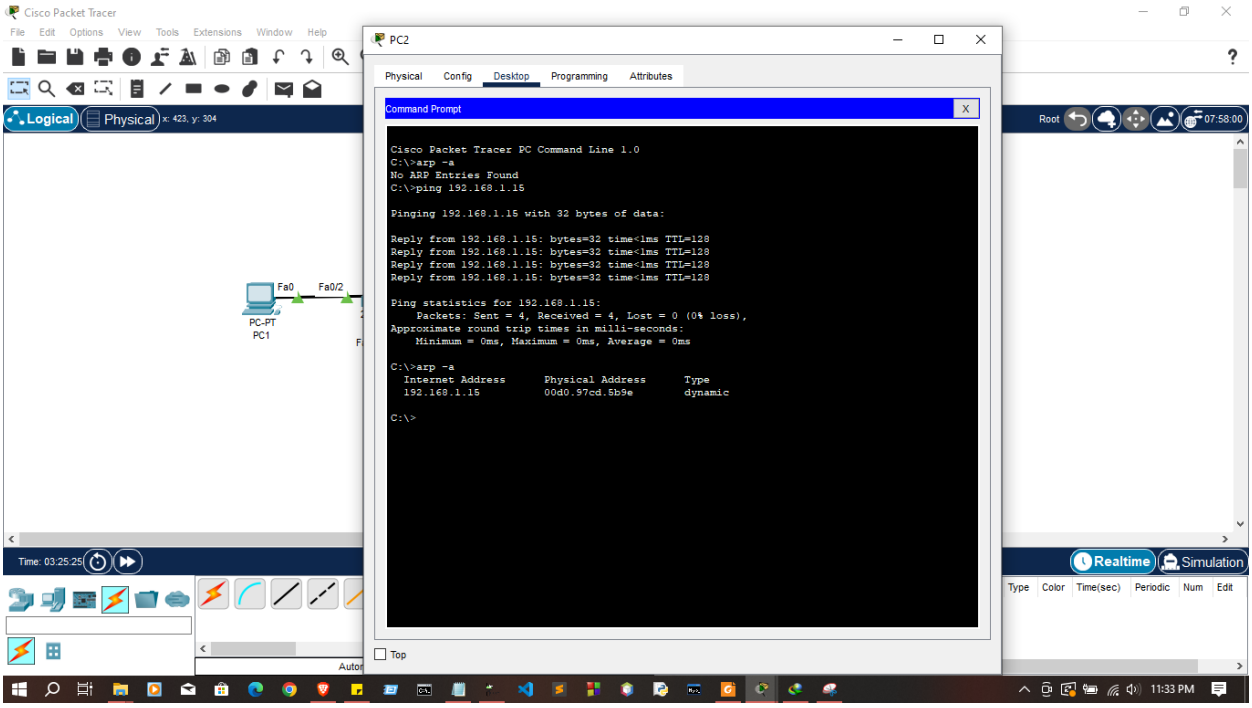
2. Analyze MAC address table
   a. Open simulation mode in packet tracer




Cisco Packet Tracer - D:\Second year frist Sem Assignment & Labsheet\03.NST21022 Practical for Network Switching and Routing\NST21022 Labsheet 03\EX01\Ex01.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Simulation Panel
Event List
Vis. Time(sec) Last Device
3.955 S1
3.955 S1
3.955 S1
3.956 S2
Reset Simulation
Constant Delay
Play Controls
If last event, capture then forward (Alt + C).
Event List Filters - Visible Events
Edit Filters
Show All/None
PC-PT PC1
PC-PT PC2
PC-PT PC3
PC-PT PC4
Fa0
Fa0/1
Fa0/2
Fa0/3
Fa0/4
Event List
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)
10:41 AM

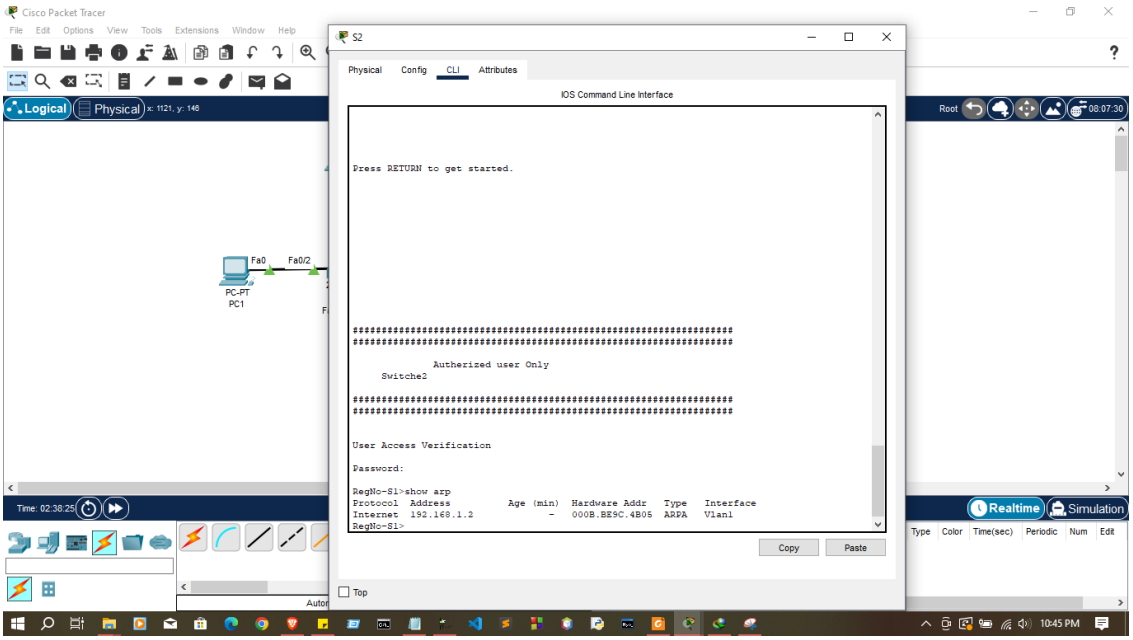

Cisco Packet Tracer - D:\Second year frist Sem Assignment & Labsheet\03.NST21022 Practical for Network Switching and Routing\NST21022 Labsheet 03\EX01\Ex01.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Simulation Panel
Event List
Vis. Time(sec) Last Device
7.950 S2
7.950 S2
7.950 S2
7.950 S2
Reset Simulation
Constant Delay
Play Controls
Event List Filters - Visible Events
Edit Filters
Show All/None
PC-PT PC1
PC-PT PC2
PC-PT PC3
PC-PT PC4
Fa0
Fa0/1
Fa0/2
Fa0/3
Fa0/4
Event List
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)
10:41 AM

b. Ty to ping from PC to another PC



c. Check ARP table in switches

show arp



RegNo-S1>
RegNo-S1>show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.1.1 - 0060.4763.DE8C ARPA Vlan1
RegNo-S1>

d. Check MAC address table in switches
show mac-address-table

User Access Verification

Password:

RegNo-S1>show mac-address-table
Mac Address Table

Vlan Mac Address Type Ports
---- ----------- -------- -----

1 0050.0f76.0162 DYNAMIC Fa0/3
1 0090.0c84.edb8 DYNAMIC Fa0/4
1 00e0.a376.e501 DYNAMIC Fa0/4
RegNo-S1>

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up


#####################################################################
#####################################################################

               Autherized user Only

#####################################################################
#####################################################################


User Access Verification

Password:

RegNo-S1>show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0050.0f76.0162    DYNAMIC     Fa0/3
   1    0090.0c84.edb8    DYNAMIC     Fa0/4
   1    00e0.a376.e501    DYNAMIC     Fa0/4
RegNo-S1>
```

Copy Paste

e. Check ARP table in PCs

arp -a

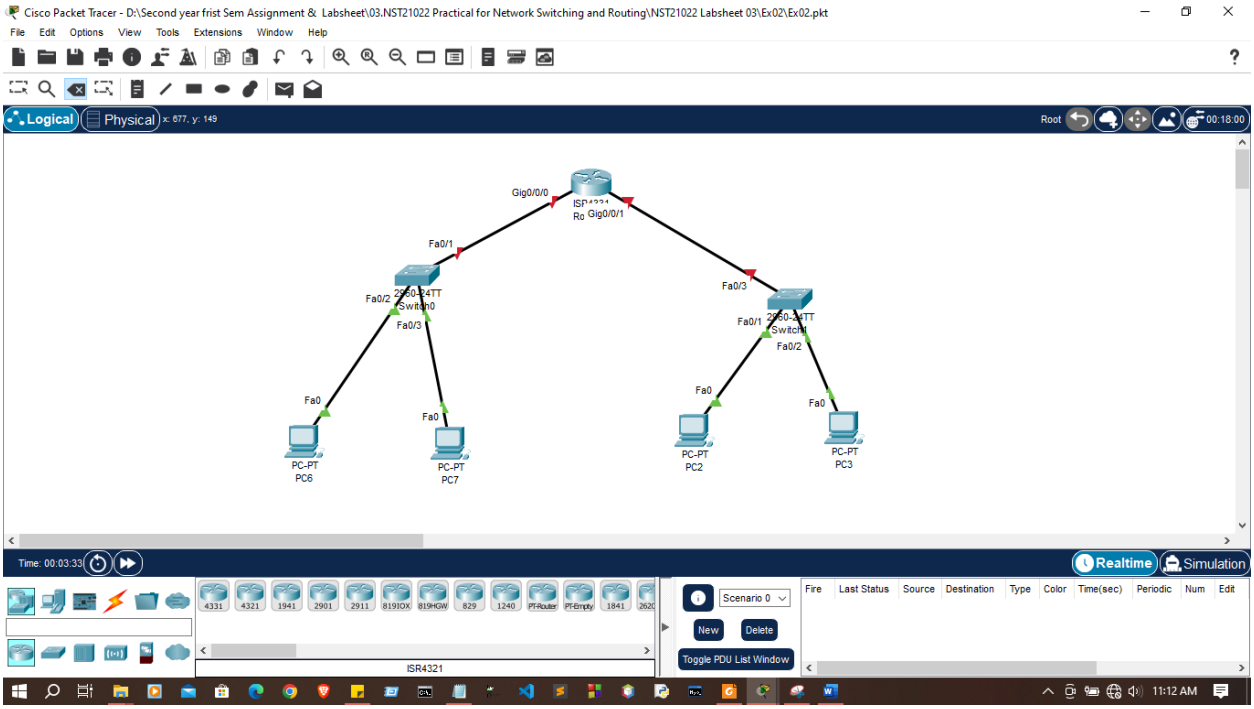
```
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.13          0090.0c84.edb8        dynamic

C:\>
```

C:\>arp -a
Internet Address Physical Address Type
192.168.1.13 0090.0c84.edb8 dynamic

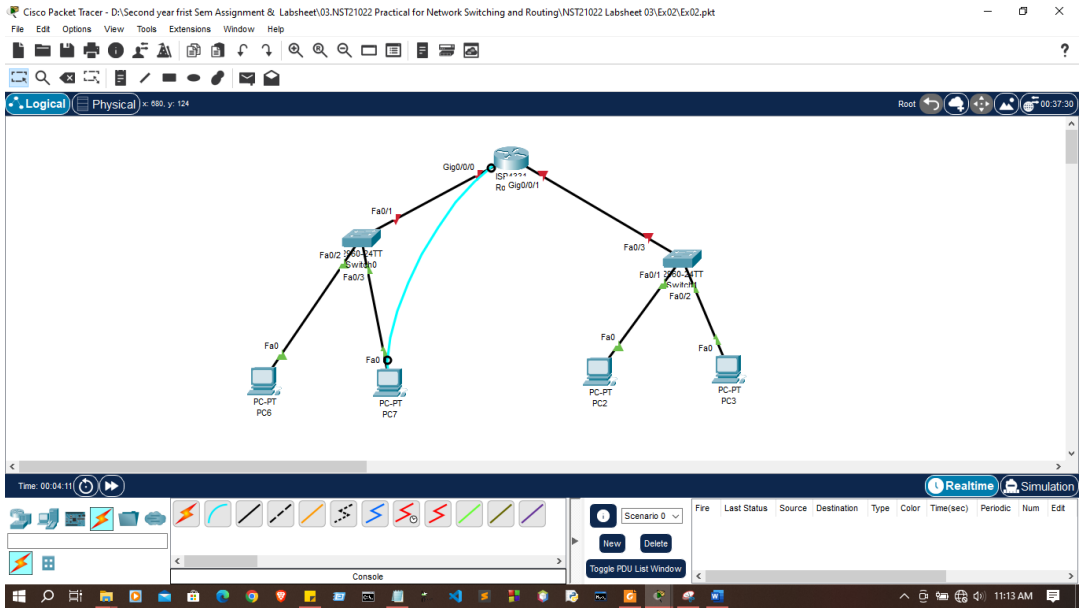
Exercise 02:

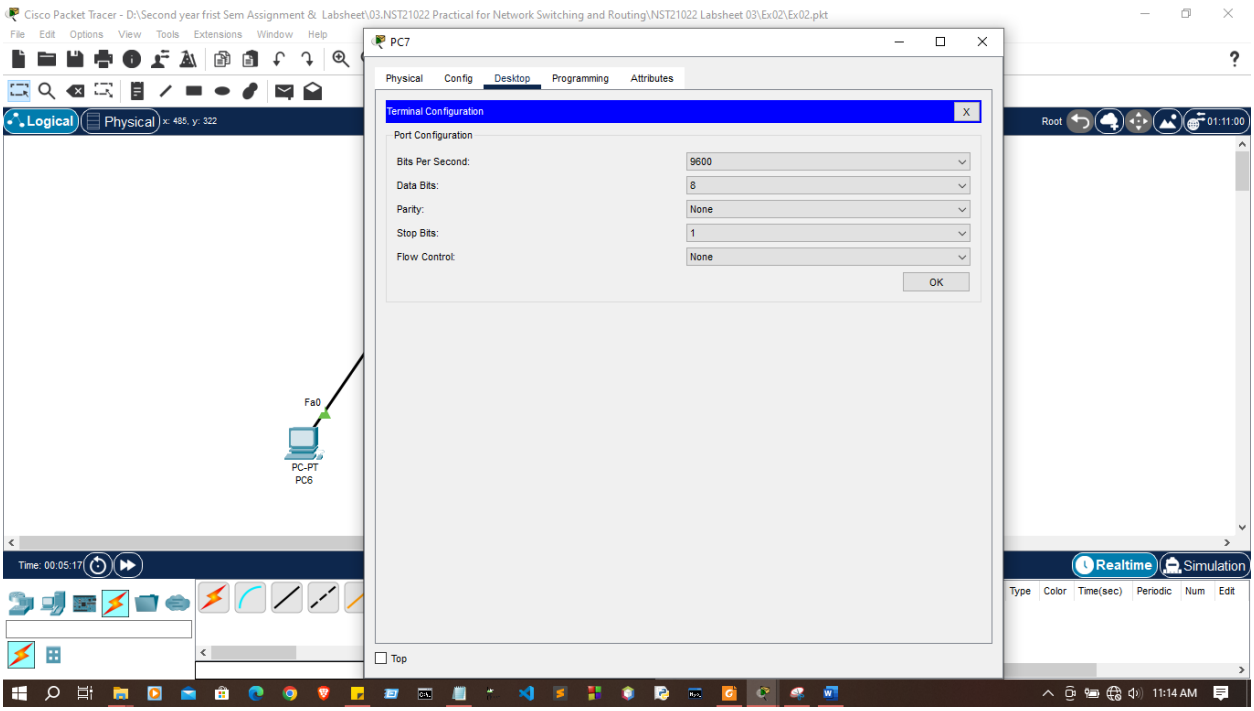
Devices	Interfaces	IP Addresses	Subnet Mask
R1	g0/0/0	192.168.1.1	255.255.255.0
	g0/0/1	192.168.2.1	255.255.255.0
PC-1	NIC	192.168.1.3	255.255.255.0
PC-2	NIC	192.168.1.4	255.255.255.0
PC-3	NIC	192.168.2.3	255.255.255.0
PC-4	NIC	192.168.2.4	255.255.255.0

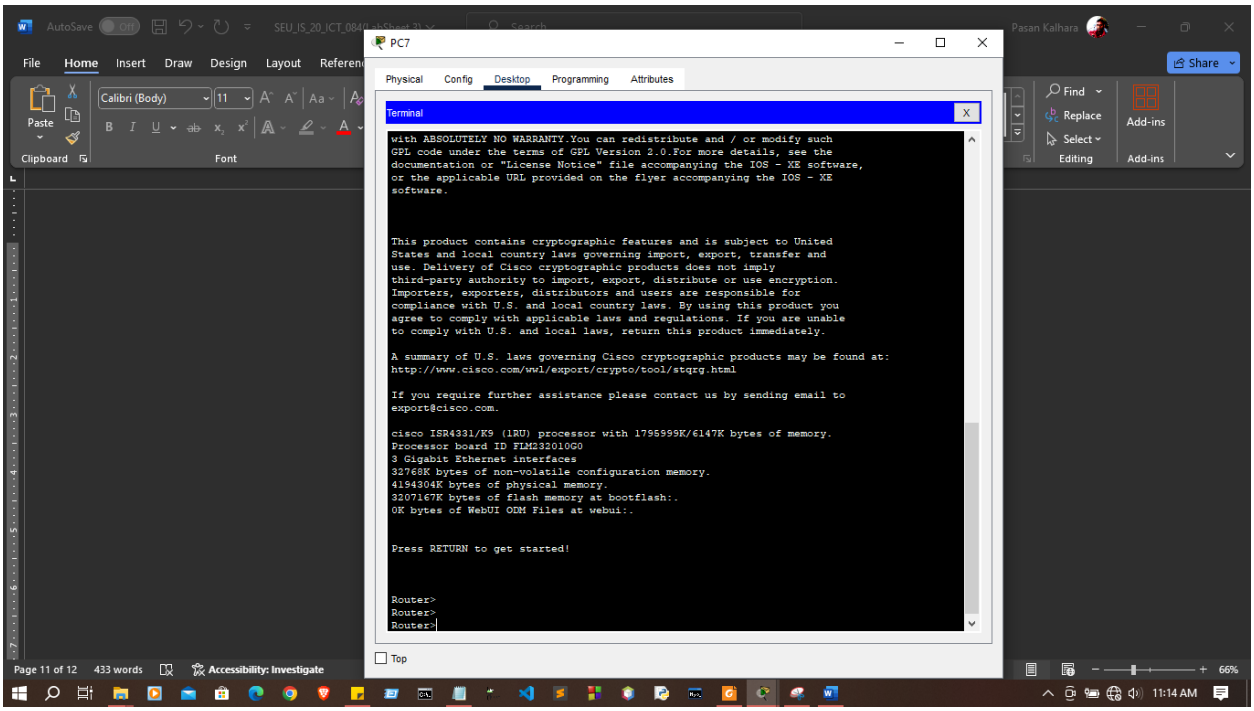
1. Configure a simple network with following devices
   a. 1 Router
   b. 2 Switches
   c. 4 PCs



2. Configure basic configuration on router through console cable.




Cisco Packet Tracer - D:\Second year frist Sem Assignment & Labsheet\03.NST21022 Practical for Network Switching and Routing\NST21022 Labsheet 03\Ex02\Ex02.pkt
PC7
Physical
Config
Desktop
Programming
Attributes
Terminal Configuration
Port Configuration
Bits Per Second: 9600
Data Bits: 8
Parity: None
Stop Bits: 1
Flow Control: None
OK
Top
Fa0
PC-PT
PC6
Logical
Physical
Realtime
Simulation


PC7
Physical
Config
Desktop
Programming
Attributes
Terminal
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GPL code under the terms of GPL Version 2.0.For more details, see the
documentation or "License Notice" file accompanying the IOS - XE software,
or the applicable URL provided on the flyer accompanying the IOS - XE
software.
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
cisco ISR4331/K9 (1RU) processor with 1795999K/6147K bytes of memory.
Processor board ID FLM232010G0
3 Gigabit Ethernet interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3207167K bytes of flash memory at bootflash:.
0K bytes of WebUI OEM Files at webui:.
Press RETURN to get started!
Router>
Router>
Router>
Top

a. Configure hostname

Router>**enable**
Router#**configure terminal**
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#**hostname R1**
R1(config)#

b. Prevent unwanted DNS lookup

```
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#
```

Top

c. Configure password protection to console line

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#line console 0
R1(config-line)#password 1234
R1(config-line)#loging
                   ^
% Invalid input detected at '^' marker.

R1(config-line)#login
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

d. Configure password protection to privileged EXEC mode

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#
R1(config)#enable secret class
R1(config)#service password-encryption
R1(config)#
```

e. Encrypt all password

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#
R1(config)#enable secret class
R1(config)#service password-encryption
R1(config)#
```

f. Assign IP addresses to the interfaces according to the addressing table

```
R1(config)#
R1(config)#
R1(config)#
R1(config)#interface g0/0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R1(config-if)#
R1(config-if)#interface g0/0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

R1(config-if)#
```

g. Configure proper banner message

```
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#banner motd $
Enter TEXT message.  End with the character '$'.

############################################################
############################################################

              Autherized user Only

############################################################
############################################################
$

R1(config)#
```

] Top

h. Save configurations

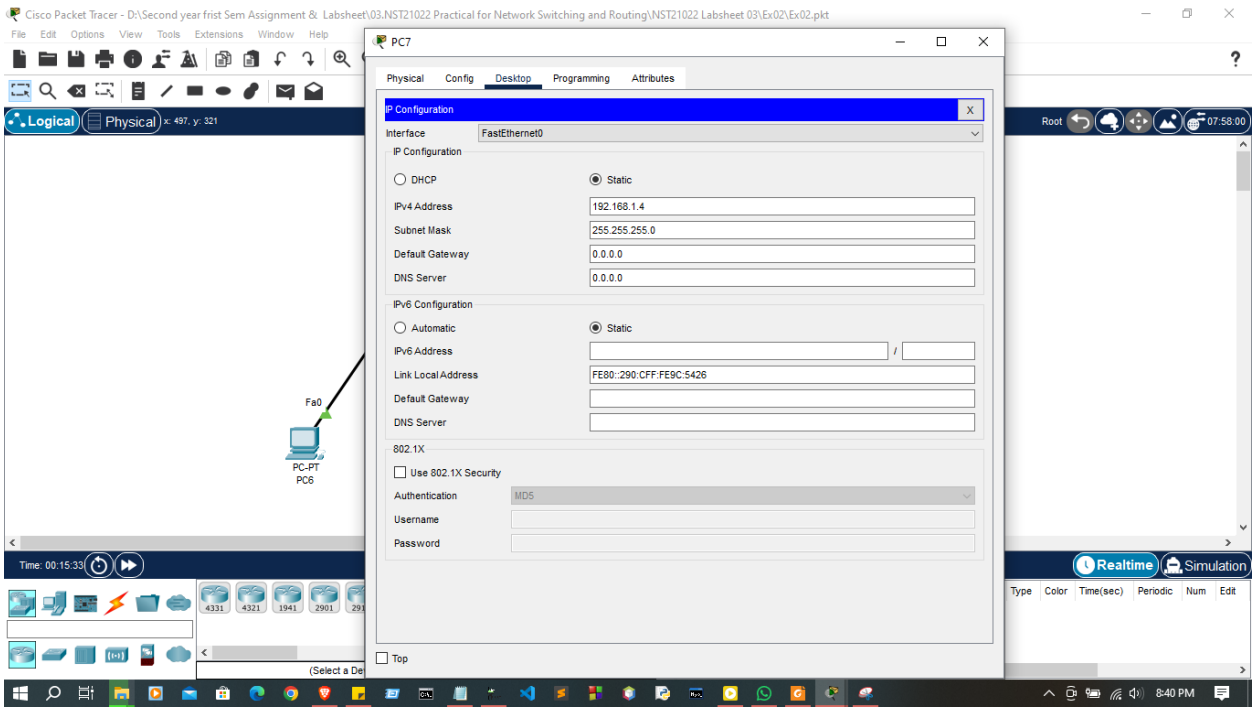
```
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
R1#
R1#
```

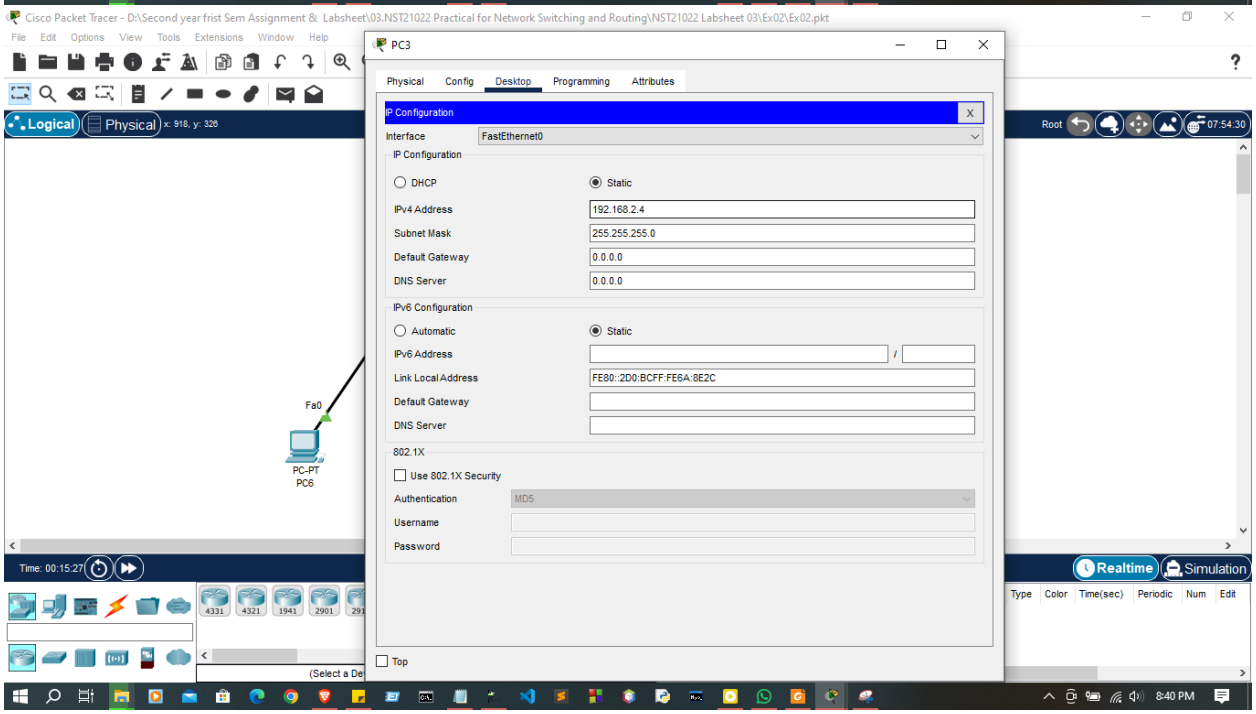
3. Verify running configurations on router

```
##################################################################
##################################################################

              Autherized user Only

##################################################################
##################################################################


User Access Verification

Password:

R1>enable
Password:
R1#show running-config
Building configuration...

Current configuration : 1084 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
```

4. Try to ping between PCs

If pinging not success discuss and find a solution


Cisco Packet Tracer - D:\Second year frist Sem Assignment & Labsheet\03.NST21022 Practical for Network Switching and Routing\NST21022 Labsheet 03\Ex02\Ex02.pkt
PC7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.4
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::290:CFF:FE9C:5426
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Fa0
PC-PT
PC6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PC3
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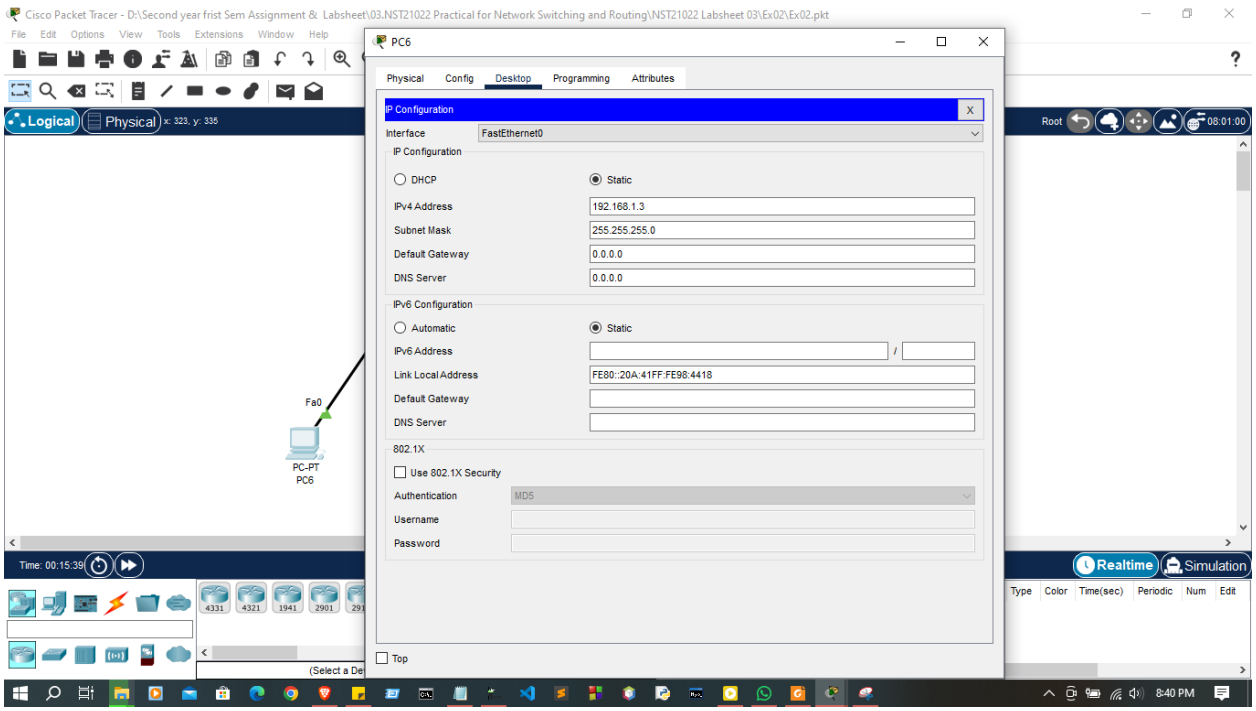
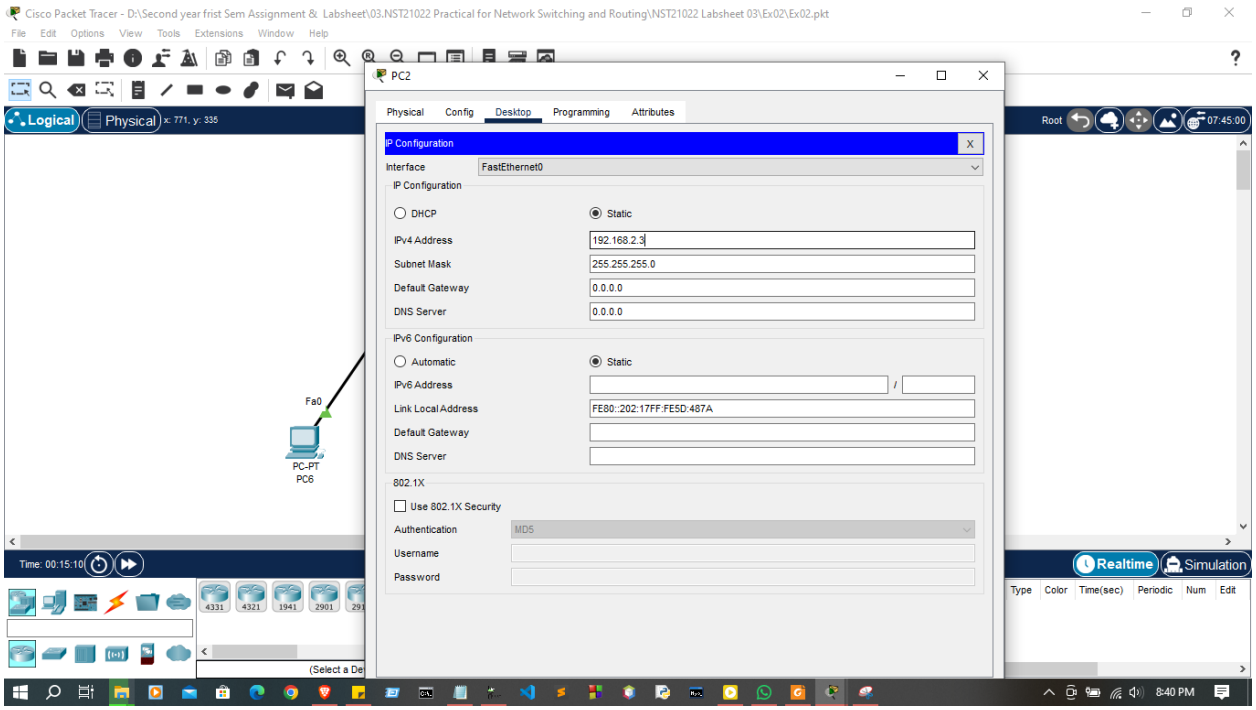
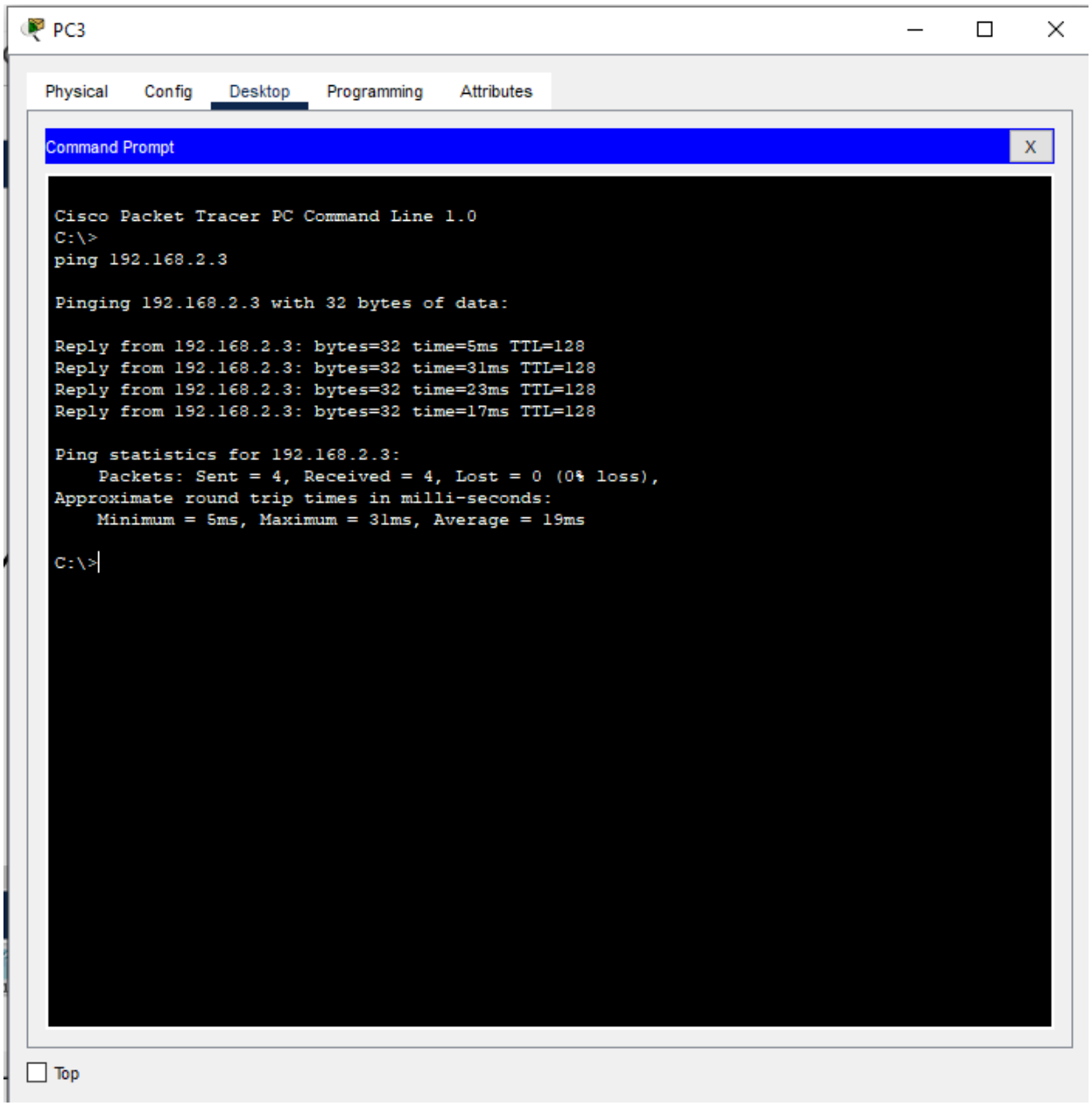
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.4
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:BCFF:FE6A:8E2C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Fa0
PC-PT
PC6


Cisco Packet Tracer - D:\Second year frist Sem Assignment & Labsheet\03.NST21022 Practical for Network Switching and Routing\NST21022 Labsheet 03\Ex02\Ex02.pkt
PC6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20A:41FF:FE98:4418
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Fa0
PC-PT
PC6
Logical
Physical
Realtime
Simulation


Cisco Packet Tracer - D:\Second year frist Sem Assignment & Labsheet\03.NST21022 Practical for Network Switching and Routing\NST21022 Labsheet 03\Ex02\Ex02.pkt
PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::202:17FF:FE5D:487A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Fa0
PC-PT
PC6
Logical
Physical
Realtime
Simulation


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time=5ms TTL=128
Reply from 192.168.2.3: bytes=32 time=31ms TTL=128
Reply from 192.168.2.3: bytes=32 time=23ms TTL=128
Reply from 192.168.2.3: bytes=32 time=17ms TTL=128
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 5ms, Maximum = 31ms, Average = 19ms
C:\>
Top

Discussion:

- In this lab session, we focused on understanding the significance of MAC addresses and routers in a network. First, we created a simple network consisting of devices like PCs and switches, and checked the MAC addresses of each device, recognizing that MAC addresses are unique identifiers for network interfaces and essential for communication at the data link layer. Next, we introduced a router to the network to facilitate communication between different networks. We configured the router with basic settings, such as IP addressing, and tested connectivity between devices. This session helped us understand the critical role of MAC addresses in device identification and how routers enable communication across different networks.